

Memo 03: MP1 Mass-Conservation Check and Updated Diagnosis

FVCOM-MP Delaware Test Cases

May 1, 2026

Purpose

This memo updates the MP1 mass-conservation diagnosis after the new case **a** test in which sediment runtime was active from the beginning of the run, but the sediment horizontal `adv_scal_backward` and `fct_sed` block in `mod_sed.F` was bypassed. The new result is the strongest evidence so far: case **a** returns to the old v001/v004-like mass behavior when that sediment advection/FCT stage is closed.

Archived Analysis Outputs

Five mass-conservation snapshots are now preserved under `FVCOM_MP_test_run/PYTHON/output/mass_conservation/archive/`.

Version	Archive label	Meaning
v001	v001_before_case_a_sediment_on	old comparison with case a using <code>SEDIMENT_MODEL=F</code>
v002	v002_case_a_sediment_on_full10d	case a with <code>SEDIMENT_MODEL=T</code> and the original sediment BBL settings
v003	v003_case_a_sediment_on_no_bbl	case- a diagnostic with <code>SEDIMENT_MODEL=T</code> and sediment BBL roughness switches disabled
v004	v004_sed_start_delayed	case a with sediment initialized but active <code>Advance_Sed</code> delayed beyond the run
v005	v005_sed_no_adv_fct	case a with active sediment runtime, but sediment <code>adv_scal_backward/fct_sed</code> bypassed

The live mass-conservation output folder has also been regenerated from the v005 state. The archived files include `mass_summary.csv`, `mass_timeseries_long.csv`, and all current comparison plots.

Case-a Evidence Trail

All archived comparisons cover the same FVCOM model-day window, 58485.0–58494.957031, or comparison day 0.00–9.96.

The two low-mass runs are v002 and v003, both of which execute active sediment horizontal advection and FCT limiting. The two high-mass controls are v001 and v004: either sediment is absent, or sediment setup is present but active `Advance_Sed` returns before runtime sediment transport. The new v005 result is the key isolation: it keeps active sediment runtime and starts sediment

Version	Case-a sediment treatment	Final total (kg)	Final vs. v001 (kg)
v001	sediment model off	1001.25	0.00
v002	sediment on, original BBL settings	582.41	-418.85
v003	sediment on, BBL disabled	573.28	-427.97
v004	sediment initialized, <code>Advance_Sed</code> delayed	1001.25	0.00
v005	sediment on, <code>adv_scal_backward/fct_sed</code> by-passed	994.20	-7.05

immediately (`SED_START` = 1), but bypasses the sediment horizontal advection/FCT block. Case a then recovers to 994.20 kg, within 7.05 kg of the old v001/v004 level.

Current v005 Output Summary

Case	Current output state	Final water (kg)	Final bed (kg)	Final total (kg)
a	rerun with sediment advection/FCT by-passed	994.20	0.00	994.20
b	previous output; not yet rerun with mirrored sediment input	371.85	0.00	371.85
c	previous output; not yet rerun with mirrored sediment input	504.26	24.29	528.55

The hard concentration cap remains inactive. The largest saved MP1 concentration in the v005 live analysis is $4.37 \times 10^{-4} \text{ kg m}^{-3}$ for case a, $6.98 \times 10^{-3} \text{ kg m}^{-3}$ for case b, and $1.45 \times 10^{-2} \text{ kg m}^{-3}$ for case c; all cap counters are zero. The saved split fields are also internally consistent: `mp1_agg` + `mp1_dis` $\approx$ `mp1`, with maximum total split-minus-combined mismatch of zero for case b and $1.83 \times 10^{-7} \text{ kg}$ for case c.

Relevant Call Trace

1. **Setup order.** In `fvcom.F:643--651`, `SETUP_SED` is called before `SETUP_PLAST`.
2. **Timestep order.** In `internal_step.F:1092--1137`, `ADVANCE_SED` is called before `ADVANCE_PLAST`. Plastic therefore sees whatever state sediment leaves behind each internal step.
3. **Sediment startup guard.** In `mod_sed.F:883`, `Advance_Sed` returns immediately when `iint` < `sed_start`. The v004 test used this to keep sediment setup active while removing active sediment runtime.
4. **Bypassed block in the current build.** In `mod_sed.F:1016--1033`, the sediment `adv_scal_backward` call and `fct_sed` limiter are commented, while `sed(i)%cnew = sed(i)%conc` and `sed(i)%cflx = 0.0_sp` keep the rest of `Advance_Sed` moving.
5. **Shared scalar machinery.** `Adv_Scal_Backward` begins at `mod_scal.F:648`. During that routine, bottom scalar-gradient arrays `PFPXB/PFPYB` are written at `mod_scal.F:911--912`. This remains a concrete shared-state candidate, although the current test only proves that the sediment `adv_scal_backward/fct_sed` stage is implicated, not which line inside the stage is responsible.

Diagnosis

1. **The problem is no longer just “sediment runtime on”.** v002 and v003 showed that active sediment runtime could move case-a final domain MP1 mass from about 1001 kg down to 573–582 kg. v004 then showed this was not caused by sediment setup, sediment namelist presence, or the sediment-enabled river forcing path, because delaying active `Advance_Sed` recovered the old mass curve.
2. **The new smoking gun is the sediment horizontal advection/FCT stage.** v005 keeps active sediment runtime, starts sediment immediately, and keeps the non-advection parts of `Advance_Sed` active. Closing only the sediment `adv_scal_backward/fct_sed` block recovers the v001/v004-like case-a result. That makes this block the leading source of the unwanted sediment/plastic interaction.
3. **The effect is probably not intended flocc physics.** Case a has `PLAST_FLOC=F`; no-floc plastic should not depend on sediment concentration, sediment size, or sediment settling velocity through the flocc-rate routines. A no-floc response to sediment horizontal transport points toward shared scalar-transport state, altered boundary/source handling, or another side effect left behind before `Advance_Plasm` runs.
4. **BBL roughness stays demoted.** The v003 BBL-off run was only 9.13 kg lower than the v002 BBL-on run, while both were more than 410 kg lower than v005. BBL roughness is not the leading explanation for the present case-a mass shift.
5. **The present-stage model should keep this sediment block closed.** For the next plastic tests, the safest controlled configuration is the current compiled source with the sediment `adv_scal_backward/ fct_sed` block disabled in `mod_sed.F`, while using identical sediment input for cases a, b, and c. Then the remaining differences should be driven by the plastic configuration rather than by different sediment input settings.
6. **The b/c outputs must be regenerated before interpreting the remaining case differences.** The current v005 live folder contains a new case-a result, but cases b and c are still the previous outputs. They should be rerun with the same disabled sediment advection/FCT build and the same sediment input now used by case a.

Next Test Setup

For the next run series, cases b and c should mirror the case-a sediment setup exactly:

1. Keep the current `mod_sed.F` build in which the sediment `adv_scal_backward/fct_sed` block is bypassed.
2. Use `SEDIMENT_MODEL=T` for all three cases.
3. Point cases a, b, and c to the same sediment input file, `mp_sediment_a_no_adv.inp`. This file starts sediment at `SED_START = 1`, keeps suspended sediment active, and keeps BBL roughness closures off.
4. Leave the plastic input files case-specific: `generic_plastic_a.inp`, `generic_plastic_b.inp`, and `generic_plastic_c.inp`. This preserves the plastic behavior being tested while removing sediment-input differences.

Focused Follow-Up Checks

1. Rerun cases `b` and `c` with the mirrored sediment input and the present disabled sediment advection/FCT build. Archive those outputs as the next version before changing code again.
2. If a narrower code diagnosis is needed, split the current bypass into two tests: re-enable `adv_scal_backward` with `fct_sed` still disabled, then test `fct_sed` separately. This separates transport side effects from limiter side effects.
3. Add a targeted shared-state checksum immediately before and after `Advance_Sed`, before `Advance_Plast`. Track `PFPXB`, `PFPYB`, `D`, `DT`, `DZ`, `KH`, `U`, `V`, boundary/source flux summaries, and the shear-stress array passed to plastic.
4. Run a save/restore test around sediment `adv_scal_backward` for candidate shared arrays. If restoring one array recovers the `v001/v004/v005`-like mass curve, the coupling mechanism is identified directly.
5. For `b/c`, reconcile the sediment post-clipping exchange path. After sediment clipping, `mod_sed.F:1099--1104` updates `sed(i)%conc`; the following MPI exchange at `mod_sed.F:1117--1122` exchanges `sed(i)%cnew`, not the clipped `sed(i)%conc`. That mismatch is a plausible source of rank-boundary noise in floc cases because floc-rate calculations read `sed(ised)%conc`.